

Lucas Castillo



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Employment

July – September 2024	Predoctoral Researcher , UNIVERSITY OF HAMBURG, Germany. Carried out research project under the supervision of Prof. Sebastian Gluth.
2023 – 2024	Research Assistant , UNIVERSITY OF WARWICK, United Kingdom. Continued development of <i>samplr</i> R Package .
2021 – 2023	Teaching Assistant , UNIVERSITY OF WARWICK, United Kingdom. Assisted teaching of two statistics modules (frequentist and Bayesian) at MSc level.
2020 – 2021	Research Assistant , UNIVERSITY OF WARWICK, United Kingdom. Started the <i>samplr</i> R Package , developed lab website , and carried out own research.

Education

2021 – 2025	Ph.D. in Psychology UNIVERSITY OF WARWICK, United Kingdom. Supervisor: Prof. Adam Sanborn Thesis title: <i>A Local Sampling Account of Random Generation</i>
2019 – 2020	M.Sc. Cognitive and Decision Sciences UNIVERSITY COLLEGE LONDON, United Kingdom Dissertation title: <i>Does newly learned causal knowledge guide the perception of temporal order?</i>
2012 – 2016	B.Sc. Psychology UNIVERSIDAD DE SEVILLA, Spain.

Research Publications

Journal Articles

- Sanborn, A. N., Zhu, J.-Q., Spicer, J., León-Villagrà, P., **Castillo, L.**, Falbén, J. K., Li, Y.-X., Tee, A., & Chater, N. (2025). Noise in Cognition: Bug or Feature? *Perspectives on Psychological Science*, 17456916241258951. <https://doi.org/10.1177/17456916241258951>
- Castillo, L.**, León-Villagrà, P., Chater, N., & Sanborn, A. N. (2024). Explaining the flaws in human random generation as local sampling with momentum. *PLOS Computational Biology*, 20(1), 1–24. <https://doi.org/10.1371/journal.pcbi.1011739>

Preprints

- Castillo, L.**, León-Villagrà, P., Falbén, J. K., Chater, N., & Sanborn, A. N. (2025a). *People Need a Few Seconds to Be Random*. https://doi.org/10.31234/osf.io/awg9j_v1
- Castillo, L.**, León-Villagrà, P., Falbén, J. K., Chater, N., & Sanborn, A. N. (2025b). *Random Generation is What Comes to Mind*. https://doi.org/10.31234/osf.io/c9xq5_v1
- Castillo, L.**, Li, Y.-X., & Sanborn, A. N. (2025). *The samplr package: A tool for modeling human cognition with sampling algorithms*. https://doi.org/10.31234/osf.io/ax8hm_v1
- Li, Y.-X., Falben, J., **Castillo, L.**, Spicer, J., Zhu, J.-Q., Qian, S., Chater, N., & Sanborn, A. N. (2025). *Does utility bias mental simulations of risky events?* https://doi.org/10.31234/osf.io/awdr7_v1

- 5 Falbén, J., **Castillo, L.**, León-Villagrà, P., Chater, N., & Sanborn, A. N. (2024). *Biased mind or biased world? Assessing the accuracy of cultural beliefs that underlie social judgments*.
<https://doi.org/10.31234/osf.io/5hdbp>
- 6 León-Villagrà, P., **Castillo, L.**, Falbén, J., Chater, N., & Sanborn, A. N. (2024). *Eliciting Beliefs with Random Generation Tasks*. <https://doi.org/10.31234/osf.io/687au>

Peer-reviewed Conference Proceedings Articles

- 1 Li, Y.-X., Falbén, J. K., **Castillo, L.**, Spicer, J., Zhu, J.-Q., Chater, N., & Sanborn, A. N. (2024). Probability, but not utility, influences repeated mental simulations of lotteries. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46. <https://escholarship.org/uc/item/2ps2t4sz>
- 2 Valentin, S., **Castillo, L.**, Sanborn, A. N., & Lucas, C. G. (2024). Distinguishing Between Process Models of Causal Learning. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46.
<https://escholarship.org/uc/item/2p43j2cw>
- 3 León-Villagrà, P., **Castillo, L.**, Chater, N., & Sanborn, A. N. (2022). Eliciting Human Beliefs using Random Generation. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 44.
<https://escholarship.org/uc/item/88k526dj>
- 4 **Castillo, L.**, León-Villagrà, P., Chater, N., & Sanborn, A. N. (2021). Local Sampling with Momentum Accounts for Human Random Sequence Generation. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 43. <https://escholarship.org/uc/item/3gz154kg>

MSc Thesis

- 1 **Castillo, L.** (2020). *Does newly learned causal knowledge guide the perception of temporal order?* [Unpublished MSc Dissertation]. University College London.

Conferences

Talks

- July 2024 **MathPsych** Characterizing People's Sampling Engines Using Random Generation
- August 2023 **SPUDM** Production rate effects on random generation elicitation
- July 2023 **MathPsych** Modelling speeded random generation as sampling for inference.
- July 2022 **Virtual MathPsych** Using sampling algorithms to explain human random generation.
- October 2021 **Computational Cognition Workshop** Human random generation as a locally-bound process.

Poster Presentations

- July 2024 **Annual Meeting of the Cognitive Science Society** People Need About Five Seconds to be Random: Autocorrelated Sampling Algorithms Can Explain Why
- November 2021 **Annual Meeting of the Psychonomic Society** Local sampling can explain human deviations from randomness.
- July 2021 **Annual Meeting of the Cognitive Science Society** Local sampling with momentum accounts for human random sequence generation.

Reviews

Since	Venue	Count
2024	Journal of Experimental Psychology: General	1
2023	Journal of Experimental Psychology: Learning, Memory, and Cognition	1
2024	Cogsci Conference	10
2023	SPUDM Conference	5

Misc.

Other Outputs

- R Packages **Castillo, L.**, Li, Y.-X., & Sanborn, A. N. (2024). *samplr*: Compare Human Performance to Sampling Algorithms. <https://CRAN.R-project.org/package=samplr>.
- Castillo, L.**, Li, Y.-X., & Sanborn, A. N. (2024). *samplrData*: Datasets from the SAMPLING Project. <https://CRAN.R-project.org/package=samplrData>.

Informal Workshops Run

- July 2025 **Using Git for Version Control**, to PhD Students at UNIVERSITY OF WARWICK, United Kingdom.
- May 2025 **Giving Good Presentations**, to PhD Students at UNIVERSITY OF WARWICK, United Kingdom.
- September 2024 **L^AT_EX Primer** to PhD Students and Postdocs at UNIVERSITY OF HAMBURG, Germany.

Other Work Experience

- Summers 2013 – 2016 **Coordinator/Director of Summer Camps** at TECS in El Puerto de Santa María, Spain.
- 2018 – 2019 **English Teacher** to adults and children at THE ENGLISH CLUB in Valladolid, Spain.
- 2016 – 2017 **English Teacher** to adults at GREENWICH ENGLISH COLLEGE in Melbourne, Australia.

Awards, Grants and Scholarships

- 2025 **Best PhD Student Paper** DEPARTMENT OF PSYCHOLOGY, UNIVERSITY OF WARWICK, United Kingdom. *Granted for Castillo et al. 2024.*
- Behaviour Spotlight Small Grant Scheme (£2000)** UNIVERSITY OF WARWICK, United Kingdom (Co-applicant). Project: *Rumination and Random Generation*
- 2024 **Behaviour Spotlight Small Grant Scheme (£2000)** UNIVERSITY OF WARWICK, United Kingdom (Co-applicant). Project: *Exploration Behaviour in Decisions from Experience*
- 2021 **Departmental PhD Scholarship** DEPARTMENT OF PSYCHOLOGY, UNIVERSITY OF WARWICK, United Kingdom
- 2020 **Best performing student** of the academic year for the MSc in Cognitive and Decision Sciences, UNIVERSITY COLLEGE LONDON, United Kingdom.

Certifications

- 2015 **Cambridge Certificate in English Language Teaching to Adults (CELTA)**. Awarded by CLIC INTERNATIONAL HOUSE in Sevilla, Spain.

Skills

- Languages Spanish (mother tongue), Catalan, English (fluent), French, German (learning!)
- Coding Python, R, L^AT_EX, C++.

Skills (continued)

Web Dev	HTML, CSS, JavaScript, REACTJS.
Databases	PostgreSQL.
Offline GUIs	Psychopy, Qt5

References

Available on Request